IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MATHEMATICS 1A

Wednesday 10 January 10:00 am

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks (25% each).

NO CALCULATORS ALLOWED.

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Second Marker: T.Pike

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ELEC40012 - MATHEMATICS1A - QUESTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. Given complex numbers $z_1 = e^{i\pi/6}$ and $z_2 = e^{-i\pi/3}$, these can be written in cartesian form,

   (a) $z_1 - z_2 = \dfrac{\sqrt{3}}{2} - \dfrac{1}{2} - \left(\dfrac{\sqrt{3}}{2} + \dfrac{1}{2}\right) i$; (b) $z_1 z_2 = \dfrac{1}{2} - \dfrac{\sqrt{3}}{2} i$;

   (c) $z_1 + z_2 = \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} + \left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right) i$; (d) $z_2/z_1 = -i$;

   (e) none of the above. [3]

2. Which of the following is the locus $z = x + iy$, in the complex plane, satisfying

   $$|z - i| = 2|i(z+1)| \ ?$$

   (a) A parabola with equation $x = -\dfrac{2}{3}y^2 + \dfrac{1}{9}y - \dfrac{4}{3}$;

   (b) a circle with centre $\left(-\dfrac{4}{3}, -\dfrac{1}{3}\right)$ and radius $\dfrac{2}{3}\sqrt{2}$;

   (c) a circle with centre $\left(\dfrac{1}{3}, \dfrac{4}{3}\right)$ and radius $\dfrac{1}{3}\sqrt{2}$;

   (d) an ellipse with equation $2(x-2)^2 + 3(y-1)^2 = 1$;

   (e) none of the above. [3]

3. Which of the following statements about $\lim_{x\to\infty} x\left(\sqrt{x^2+4} - x\right)$ is true?

   (a) The limit is equal to 0 ; (b) the limit does not exist ;

   (c) the limit is equal to 2 ; (d) the limit cannot be determined ;

   (e) none of the above. [3]

4. The limit $\lim_{x\to 0} \dfrac{x^2}{\ln(\cos x)}$

(a) exists, but cannot be determined, ; (b) the limit does not exist ;

(c) is equal to -2 ; (d) is equal to 0 ;

(e) none of the above. [3]

5. The graph of the function $y = x^5 - 5x^4$ has

(a) a point of inflection at $x = 0$, and a local minimum at $x = 4$;

(b) points of inflection at $x = 0$ and $x = 3$;

(c) a point of inflection at $x = 3$, and a local maximum at $x = 0$;

(d) a point of inflection at $x = 3$, and a local minimum at $x = 0$;

(e) none of the above. [3]

6. If $\cos y = e^{-x}$, for $x \geq 0$, which of the following is $\dfrac{dy}{dx}$?

(a) $\cot y$; (b) $-\dfrac{e^{-x}}{\sqrt{1-e^{-2x}}}$; (c) $\tan y$; (d) $\dfrac{e^{-2x}}{\sqrt{1-e^{-x}}}$;

(e) none of the above. [2]

7. If $y = \tan u$, where $u = v + \dfrac{1}{v}$ and $v = \ln x$, what is the value of $\dfrac{dy}{dx}$ at $x = e$?

(a) $\dfrac{1}{e}$; (b) -1 ; (c) $\dfrac{2}{e^2}$; (d) $\sec^2(e)$;

(e) none of the above. [3]

8. If $n = 0, 1, 2, 3, \ldots$, for which values of n is it true that

$$\int_0^1 x^n \, dx \;=\; \int_0^1 (1-x)^n \, dx \,?$$

(a) all of them ; (b) even values ; (c) odd values ; (d) none of them ;

(e) none of the above . [2]

9. The integral $I = \displaystyle\int \frac{1}{1-\sin x}\, dx$ is equal to

(a) $\dfrac{2}{1-\tan(x/2)} + \mathrm{C}$; (b) $\cot x + \sec x + \mathrm{C}$; (c) $\dfrac{x}{1+\cos x} + \mathrm{C}$; (d) $\dfrac{2}{1+\cot(x/2)} + \mathrm{C}$;

(e) none of the above. [3]

10. Two series are given as

$$S_1 = \sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{and} \quad S_2 = \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}} \, .$$

Which of the following statements is true?

(a) S_2 diverges, and S_1 has infinite radius of convergence;

(b) S_2 converges, and S_1 has radius of convergence $R = 1$;

(c) S_2 has radius of convergence $R = 1$, and S_1 diverges for $|x| > 1$;

(d) S_2 diverges, and S_1 has interval of convergence $[-1, 1)$;

(e) none of the above. [3]

11. The Maclaurin series for $f(x) = \sqrt{1-x}$ to order two, with Lagrange remainder term is given by

$$1 - \frac{1}{2}x - \frac{1}{8}x^2 + R_2 \, .$$

Which of the following are the correct Lagrange remainder, and inequality regarding the uncertainty expressed in c?

(a) $R_2 = -\frac{x^3}{16(1-c)^{5/2}}$ and $0 < |c| < |x|$;

(b) $R_2 = -\frac{x^2}{8(1-c)^{3/2}}$ and $0 < c < x$;

(c) $R_2 = \left| \frac{x^3}{16(1-c)^{5/2}} \right|$ and $0 < |c| < |x|$;

(d $R_2 = -\frac{x^3}{16(1-x)^{5/2}}$ and $0 < c < x$;

(e) none of the above. [3]

12. The first order ordinary differential equation (ODE)

$$xy' = \frac{1}{x}(y-1)$$

has general solution

(a) $y = Ce^{1/x} - 1$; (b) $y = 1 + Ce^{-1/x}$; (c) $y = 1 - e^{-1/x}$+C ; (d) $y = \dfrac{1}{x + Ce^{-1/x}}$;

(e) none of the above . [3]

13. Given the linear second-order ODE

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 2x - 1 \, ,$$

which of the following is the complementary function?

(a) $y_c = c_1 x e^{-x} + c_2 e^{2x}$; (b) $y_c = c_1 e^{-x} + c_2 e^{2x}$;

(c) $y_c = c_1 e^{-x} + c_2 e^{-2x}$; (d) $y_c = c_1 e^{-2x} + c_2 x e^{x}$;

(e) none of the above. [2]

14. Given the linear second-order ODE

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 2x - 1\,,$$

which of the following is a particular integral?

(a) $y_p = -x^2 + x - \frac{3}{2}$; (b) $y_p = e^{-x} + e^{2x}$; (c) $y_p = x^2 - x - \frac{1}{2}$; (d) $y_p = -x^2 - \frac{1}{2}$;

(e) none of the above. [2]

15. Given the linear second-order ODE

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 2x - 1\,,$$

which of the following is the solution satisfying $y = 1$ and $y' = 0$ when $x = 0$?

(a) $y = -xe^{-x} - \frac{1}{2}e^{2x} - x^2 + x - \frac{3}{2}$; (b) $y = \frac{1}{2}e^{-x} - e^{-2x} + x^2 - x - \frac{1}{2}$;

(c) $y = e^{-x} + \frac{1}{2}e^{2x} - x^2 - \frac{1}{2}$; (d) $y = -\frac{1}{2}e^{-2x} + xe^{x} + e^{-x} + e^{2x}$;

(e) none of the above. [2]

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. a) For $z = x + iy$, use the exponential definition of the hyperbolic cosine function to obtain $\cosh(iz)$ in the form $A + Bi$, where A, B are real trigonometric and hyperbolic functions.

Hence, or otherwise, show that [7]

$$|\cosh(iz)|^2 = \frac{1}{2}\cosh 2y + \frac{1}{2}\cos 2x\,.$$

b) Given the function

$$f(x) = \frac{1}{1+e^{-x}}\,,$$

obtain $f_e(x)$, the even part of $f(x)$. With no further calculation, write down the odd part, $f_o(x)$, and explain your reasoning. Hence, use these to sketch $f(x)$. [6]

c) If y is a function of x, prove the inverse function rule $\dfrac{dx}{dy} = \dfrac{1}{\frac{dy}{dx}}$,

and obtain the corresponding expression for $\dfrac{d^2x}{dy^2}$ in terms of derivatives of y with respect to x. [7]

2. a) Given the limit

$$L = \lim_{x\to\infty} x^q \left[(2x+1)^{1/2} - (2x-1)^{1/2}\right]$$

find the value q so that L exists, and is non-zero. Also calculate L. [6]

b) Given the function $f(x) = (x^2 - x - 2)e^{2x}$, obtain all stationary points and use the second derivative to classify them. Find all intercepts, points of inflection and asymptotes, and sketch the function, explaining your results. [6]

c) Given the integral

$$I_n = \int \frac{1}{(1-x^2)^n}\, dx,$$

show that

$$I_n = I_{n-1} + \int \frac{x^2}{(1-x^2)^n}\, dx\ .$$

Now integrate

$$\int \frac{x^2}{(1-x^2)^n}\, dx$$

by parts, and find an integral recurrence relation for I_n in terms of I_{n-1} and a function of x. [8]

3. a) Calculate the integral

$$I = \int \frac{1}{1-\cos x-\sin x}\, dx\,.$$

[5]

b) Obtain the general solution of the first-order ODE [6]

$$y'(x) = \frac{x^2}{y^2} + \frac{2y}{x}\,.$$

c) We seek a series solution of the second-order linear ODE

$$(1+x^2)\frac{d^2y}{dx^2} + \frac{3}{2}x\frac{dy}{dx} - \frac{1}{2}y = 0$$

satisfying the initial conditions $y = 1$ and $y' = 0$, when $t = 0$.

(i) Use Leibnitz' theorem to differentiate the ODE n times. [3]

(ii) Let $x = 0$ in your result from (i) to obtain a recurrence relation

$$y^{(n+2)}(0) = f(n)y^{(n)}(0)\,, \qquad (n \geq 0)$$

where $f(n)$ needs to be determined, and $y^{(k)}(0)$ is y differentiated k times, and evaluated at $x = 0$. [3]

(iii) Use the recurrence relation and initial conditions to obtain the first three non-zero terms of the Maclaurin series of $y(x)$. [3]

MATHEMATICAL FORMULAE

1 Vector Algebra

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \;=\; (a_1, a_2, a_3)$$

Scalar(Dot) Product $\quad \mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

Vector (Cross) Product

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Triple vector product $\quad \mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$

Triple scalar product

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1$$

2 Series

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots+\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^r+\ldots \;(n\,\text{arbitrary}\,, |x|<1)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^r}{r!} + \ldots \qquad \text{for all } x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{r+1}\frac{x^r}{r} + \ldots \qquad \text{for } -1 < x \leq 1$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + \frac{(-1)^r x^{2r}}{(2r)!} + \ldots \qquad \text{for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \ldots \qquad \text{for all } x$$

3 Trigonometric Identities and Hyperbolic Functions

$$\sin(A+B) = \sin A\cos B + \sin B\cos A\,, \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(iz) = \cosh z\,, \quad \cosh(iz) = \cos z\,, \quad \sin(iz) = i\,\sinh(z)\,, \quad \sinh(iz) = i\,\sin z$$

$$e^{ix} = \cos(x) + i\sin(x)$$

$$\cos(x)\cos(y) = \frac{1}{2}\cos(x-y) + \frac{1}{2}\cos(x+y)$$

$$a\cos x + b\sin(x) = c\cos(x+\theta) \quad \text{where} \quad c = \sqrt{a^2+b^2} \quad \text{and} \quad \theta = \tan^{-1}\left(-\frac{b}{a}\right)$$

4 Differential Calculus

1. Leibnitz' Formula: $D^n(fg) =$

$$D^n f\, g + \binom{n}{1} D^{n-1} f\, Dg + \binom{n}{2} D^{n-2} f\, D^2 g + \cdots + \binom{n}{r} D^{n-r} f\, D^r g + \cdots + f D^n g$$

where $\left(\begin{array}{c} n \\ r \end{array} \right) = \dfrac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$.

2. Taylor expansion of $f(x)$ about the point $x = a$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where

$$R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c), \qquad a < c < x$$

Putting $x = a + h$ gives the form

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \ldots$$

3. Taylor expansion of $f(x, y)$ about the point (a, b):

$$f(x,y) = f(a,b) + \frac{1}{1!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right] f(a,b) + \frac{1}{2!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right]^2 f(a,b) + \cdots$$

4. Chain rule for partial derivatives: $f = f(x, y)$

(i) If $x = x(t)$ and $y = y(t)$ then $f = F(t)$ and

$$\frac{dF}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

(ii) If x and y are expressed in terms of new variables u and v by $x(u, v)$ and $y(u, v)$ then $f = F(u, v)$ and

$$\frac{\partial F}{\partial u} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial u}, \quad \text{and} \quad \frac{\partial F}{\partial v} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial v}$$

(iii) If $y = y(x)$ then $f = F(x)$ and

$$\frac{dF}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx}$$

5. Stationary points of $f(x, y)$ occur where $f_x = 0$ and $f_y = 0$. Let (a, b) be a stationary point, and consider $D = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$. Then

(a) If $D > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a maximum;

(b) If $D > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a minimum;

(c) If $D < 0$ then (a, b) is a saddle point;

(d) if $D = 0$, more information is required.

6. Differential Equations

(i) The first order linear equation $\dfrac{dy}{dx} + p(x)y = r(x)$ has an integrating factor $I(x) = \exp\left\{\displaystyle\int p(x)\, dx\right\}$ which allows one to write the DE in the form $\dfrac{d}{dx}(I(x)y) = I(x)r(x)$.

(ii) The first-order equation $P(x, y)\dfrac{dy}{dx} + Q(x, y) = 0$ is exact if $\dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y}$.

5 Integral Calculus

1. An important substitution: $\tan(\theta/2) = t$; then

$$\sin\theta = \frac{2t}{1+t^2}\,, \quad \cos\theta = \frac{1-t^2}{1+t^2}\,, \quad d\theta = \frac{2\ dt}{1+t^2}\,.$$

2.

$$\int \frac{dx}{(a^2-x^2)^{1/2}} \;=\; \sin^{-1}\left(\frac{x}{a}\right)\,, \quad |x| < a\,.$$

3.

$$\int \frac{dx}{(a^2+x^2)^{1/2}} \;=\; \sinh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}+1\right)^{1/2}\right]\,.$$

4.

$$\int \frac{dx}{(x^2-a^2)^{1/2}} \;=\; \cosh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}-1\right)^{1/2}\right]\,.$$

5.

$$\int \frac{dx}{a^2+x^2} \;=\; \left(\frac{1}{a}\right)\tan^{-1}\left(\frac{x}{a}\right)\,.$$